



SECP2613 SYSTEM ANALYSIS AND DESIGN

SEMESTER 2 (2023/2024)

SECTION 02

ASSIGNMENT 2

TOPIC 2: PROJECT MANAGEMENT

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Question 1

- a) Based on the systems project Ahmad proposes, the feasibility can be assessed by analyzing the performance of the current system and any problems that may present. It is necessary to analyze the performance of the current system and any problems that may be present. This involves assessing user satisfaction, reliability of the system, and input and output procedures. Moreover, a new system's possible advantages and disadvantages should be evaluated considering possible costs and disruptions.
- b) According to Abu's statements about the managers, users, and system administrators, the suggested project's operational feasibility appears to be relatively high. Even though Abu's concern about the disruptive impacts of a new system such as user training and system integration, it is not a big problem, it can be solved. This is because the most important thing is the systems team certified that the technology to update the system is available. Moreover, users and management have indicated satisfaction with the current system, as it meets their operating requirements well.
- c) Regarding the economic feasibility, Abu points out that the new system change will require a significant investment (“big bucks for a new gimmick”), indicating the uncertainty around the new system change’s sustainability. They must, for instance, allocate funds for the systems analysts, the price of the system studies, and also the staff salaries. In addition, the economic feasibility study should account for potential long-term costs including maintenance and training fees.
- d) The organization's ability to implement the suggested system feasibility is high. This is because Abu points out that the idea might be technically possible because the technology needed to update the system is already available. Moreover, Ahmad's encouragement for a new system suggests that the organization has or is ready to allocate the required resources, such as skills, hardware, and software technology, to develop the new system.

- e) Based on Ahmad and Abu's conversations and the possible influence of the planned project on the organization, it is recommended that a full-blown systems analysis be conducted. This thorough investigation will provide a better knowledge of the operational, economic, and technological implications of deploying a new system. Through an extensive evaluation, the company can learn important information about the possible advantages, dangers, and difficulties related to the project proposal.

Question 2

Phase	Activity	Detailed Activity	Weeks Required	Predecessor Task
1. Planning	Project initiation and scope definition	Define project objectives	2	-
		Resource and time estimation		
2. Analysis	Data gathering	Gather the feedback on current registration system	2	1
		Specify non-functional requirements	1	1
	Feasibility analysis	Cost-benefit analysis	2	1
		Operational, economic and technological feasibility		
3. Design	System design and development	Database design and model development	4	2
		User interface design development		
		Prototype development and user testing		
	Testing and quality assurance	Test system development and validation of the system		
4. Development		Training material development	2	3

	Training and change management			
		Change management strategy implementation		
5. Testing	System and performance testing	Validate the entire software	3	4
		Evaluate software's performance		
6. Deployment	Deployment of the system	Conduct user training	2	5
		Prepare deployment packages and documentation		
7. Maintenance	Bug fixes and updates	Track and manage software changes effectively	Ongoing	6
	Performance monitoring	Identify improvement		

Question 3

Title: Managing project within team members

Introduction:

During managing projects, in order to make the work more efficient the manager plays such a big role in what kind of System Development Life Cycle (SDLC) they want to use. For example, whether they want to use the traditional method, such as Waterfall methodology or the most popular ones, Agile methodology. Each methodology includes 6 phases; planning, analysis, design, development, implementation, and maintenance. Therefore, this report will talk about the key difference between traditional methodology and agile methodology.

Key differences:

- Flexibility
For traditional methodology, it is less flexible. This is because everything will be decided at phase 1 and it has to follow the 6 phases linearly. Therefore, it can't be changed after each phase has been completed. Meanwhile, agile methodology is more flexible. This is because agile develops the team to be adaptive to the changes during the development process as what the stakeholders need.
- Communication
For traditional methodology, the communication is formal. For example, it includes a specific communication plans and progress reports that will be distributed with many stakeholders. Meanwhile, for agile methodology, the communication is more informal. For example, agile methodology will have a regular interaction between the management team and the stakeholders.
- Timeline of works
For traditional methodology, the timeline of works will take much longer times. This is because all the needs must be approved by the stakeholders before the development can begin. Meanwhile, the timeline of works using agile methodology can be a lot faster. This is because it has iterative development cycles.
- Roles
For traditional methodology, it offers roles to project team members based on particular activities and responsibilities. Meanwhile, for agile methodology, it encourages the team members to work together on different aspects of the project throughout time, as it is resulting in a more self-organizing team structure.

Conclusion:

To conclude this report, it indirectly shows that agile methodology should be used in developing a project instead of using the traditional ones as the team members scattered across different time zones and different physical locations. Therefore, agile methodology is more recommended to use as it has more advantages than the traditional methodology.

Question 4

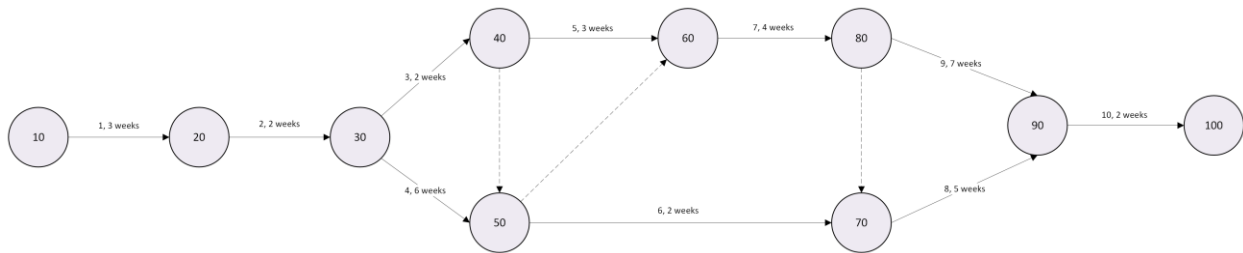
a)

Costs (RM)	Year 0	Year 1	Year 2	Year 3
Development Cost				
Hardware	50000			
Software	15000			
Training	30000			
Consulting	80000			
Data Conversion	40000			
Total Development Costs	215000			
Production Cost				
Supplies		15000	15000	15000
IS Salaries		75000	82500	90750
Upgrades		10000	10000	10000
Total Production Costs		100000	107500	115750
Present Value		74075	58985	47046
Accumulated Costs		289075	348059	395105
Benefits				
Customer Service		150000	180000	216000
Productivity		150000	187500	234375
Total Benefits		300000	367500	450375
Present Value		222223	201647	183052
Accumulated Benefits		222223	423869	606920
Gain or Loss (NPV)		66852	75810	211816
Profitability Index (PI)	0.99			

b) Since the Profitability Index is less than 1, it shows that the costs outweigh the benefits of the project. Therefore, it is not economically feasible to invest in the SkorBestari project as it is not profitable.

Question 5

a)



b)

- 1-2-3-5-7-9-10 (Path 1)
- 1-2-3-5-7-8-10 (Path 2)
- 1-2-3-6-8-10 (Path 3)
- 1-2-4-7-9-10 (Path 4)
- 1-2-4-7-8-10 (Path 5)
- 1-2-4-6-8-10 (Path 6)

c)

- Path 1: $3+2+2+3+4+7+2 = 23$ weeks
- Path 2: $3+2+2+3+4+5+2 = 21$ weeks
- Path 3: $3+2+2+2+5+2 = 16$ weeks
- Path 4: $3+2+6+4+7+2 = 24$ weeks
- Path 5: $3+2+6+4+5+2 = 22$ weeks
- Path 6: $3+2+6+2+5+2 = 20$ weeks

The critical path is Path 4 which is 1-2-4-7-9-10 with duration of 24 weeks (about 5 and a half months). The activities in this path are collecting requirements, analyzing processes, designing processes, designing reports, test and documentation and install.

d)

